Dreycey Albin, PhD

Machine Learning Engineer

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Professional Summary

ML engineer bridging applied research and production ML systems. Technical lead on ML projects for Resource Central, Microsoft Azure's central control-plane ML system, where my work across modeling, experimentation, distributed services, and low-latency inference has driven over $100M in annual capacity savings.

Technical Skills

ML: PyTorch, scikit-learn, LightGBM/XGBoost; tabular and time-series modeling; uncertainty, calibration, and validation
MLOps: Experiment tracking (Weights & Biases, MLflow); offline-to-online validation; monitoring and drift detection; CI/CD
Systems: Distributed systems; low-latency model serving and inference (ONNX); Docker, Kubernetes; Azure, AWS; Spark; Linux, Git
Languages: Python, C++, C#, Rust, SQL

Work Experience

Microsoft Azure — **Redmond, WA**

Machine Learning Engineer, Level 62 — 03/2025 – Present

- **Technical Lead, Capacity Mitigation System ($100M/yr CapEx + org-wide Awards):** Cut regional capacity mitigation from ~1 week to ~4 hours across 50+ regions and ~1M VMs/containers with zero customer impact by architecting a telemetry-driven system for capacity savings and mentoring an engineer through end-to-end delivery. *Earned Azure Core Impact and Builders Excellence awards.*
- **Technical Lead, Heterogeneous Overprovisioning Engine ($350M/yr projected):** Led a 4-person cross-functional team building ML-driven reclaim decisions for overprovisioned compute across Azure's fleet; developed a risk-adjusted LightGBM quantile regression model with offline-to-online validation, reducing under-prediction rates to less than 1% and integrating inference into critical C# control-plane services with a ≤10 ms SLO.
- **Distributed Model Delivery & Rollout Architecture (patent filed):** Architected a distributed system for automated A/B and shadow model evaluation across microservices, with canary releases, versioning, and auto-promote/rollback guardrails. Designed and prototyped during a two-week sprint, then *adopted as a broader team initiative and selected by the Microsoft patent office for filing.*

Machine Learning Engineer, Level 61 — 07/2023 – 03/2025

- **VM-SKU Policy Changes ($50M+/yr CapEx):** Led platform-wide decisioning policy changes and built the automated validation and reporting framework including forecasting, regression checks, drift monitoring, and decision stability analysis used by finance and senior leadership to track impact and control operational risk.
- **LightGBM Production Model Improvement:** Overhauled evaluation methodology with thresholding tradeoff analysis and backtesting across cohorts and edge cases; reduced false negatives by 5% while preserving safety constraints and unlocking additional capacity headroom.

Medtronic — **Boulder, CO**

Research Engineer (Contract) — 09/2021 – 05/2022

- **Real-Time LSTM Catheter Pose Estimator (<20ms inference):** Developed an LSTM model from fiber-optic sensor streams for real-time surgical catheter pose estimation with evaluation harnesses covering latency/accuracy tradeoffs for edge deployment.
- **Autonomous Catheter Robot Software Stack:** Shipped full Python software stack for an autonomous catheter robot including clinical-facing GUI, real-time motor control, telemetry, and fail-safe behaviors; helped secure additional internal R&D funding.

Education

Ph.D. Computer Science | *University of Colorado Boulder* | NSF GRFP Fellow	2020 – 2023
M.Sc. Computational Biology (SSPB Program) | *Rice University*	2018 – 2020
NIH-PREP Postbaccalaureate | *University of Washington*	2017 – 2018
B.S. Chemistry + B.S. Biology | *University of Northern Colorado* | McNair Scholar	2012 – 2017

Selected Publications

P1 **PhageScanner: a reconfigurable machine learning framework for bacteriophage genomic and metagenomic feature annotation** - (*Frontiers in Microbiology, 2024*) - D. Albin, M. Ramsahoye, E. Kochavi, M. Alistar

P2 **PhageBox: an open source digital microfluidic extension with applications for phage discovery** - (*IEEE Transactions on Biomedical Engineering, 2023*) - D. Albin, L. Buecherl, E. Kochavi, et al.

P3 **SeqScreen: a biocuration platform for robust taxonomic and biological process characterization of nucleic acid sequences of interest** - (*IEEE BIBM, 2019*) - D. Albin, D. Nasko, R.A.L. Elworth, et al.

Awards and Honors

Azure Core Impact Award	2025
Microsoft Azure Builders Excellence Award	2025
NSF Graduate Research Fellowship (GRFP)	2018 – 2023
Dissertation Completion Fellowship (Link)	2023
Outstanding Service Award, CU Boulder CS Dept. (Link)	2023